User Guide

Variables, methods, and map interpretation.

The Heat Risk Explorer is an interactive heat-risk map of San Diego County, computed at the census-tract level. Select the variables you want included (heat exposure, demographics, chronic illness, AC access, tree cover, income, and others) and the tool returns a composite score and the dominant driver for each tract.

The tool runs entirely in the browser. Nothing is installed; nothing is sent to a server.

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Quickstart

1. Open the map at connorjmack.github.io/heat-risk-viewer.
2. Leave the default variable selections.
3. Click **Build Composite Risk Map**.
4. Dark red tracts have the highest composite risk. Click any tract for its scores and dominant driver.

What goes into the score

The score combines three components: heat **exposure** (the physical heat a tract receives), population **sensitivity** (who lives there and how susceptible they are), and **adaptive capacity** (resources that reduce harm). Each component draws on multiple variables, and you choose which variables to include.

Exposure red

- **Daytime highs:** summer mean, hottest 10% of summer days, annual maximum.
- **Nighttime lows:** summer minimum. Warm nights are associated with elevated heat-illness risk in older adults.
- **Humidity:** summer humidity. High humidity reduces evaporative cooling and raises apparent temperature.

Sensitivity orange

- **Age:** share of residents under 18 and over 65.
- **Chronic conditions:** heart disease, asthma, COPD, diabetes, hypertension, and other heat-aggravated conditions (CDC PLACES).
- **Social vulnerability:** language barrier, disability, poverty, vehicle access, housing-cost burden.
- **Air quality and life expectancy:** baseline environmental and health context.

Adaptive capacity teal

- **Air conditioning:** share of homes with any AC, with central AC, or with none.
- **Built environment:** tree cover, park access, green space, housing age.
- **Resources and services:** income, education, health insurance, healthcare access, food access, community connectedness.

Higher adaptive capacity lowers the composite score.

How the components combine

The three components are combined into a single 0–100 percentile score per tract.

Correlated variables are not double-counted. Lower-income tracts tend to have older housing, less tree cover, and higher chronic-illness rates. A direct sum would weight that shared pattern multiple times. The tool uses Principal Component Analysis (PCA) to identify the underlying patterns and weight each one once.

Adaptive-capacity variables enter the score with the opposite sign. A tract with strong protective resources receives a lower composite score even when its exposure is high.

For a visual walkthrough of PCA, see this Setosa interactive.

Reading the map

Colors

Darker is higher. The composite map uses red (highest = darkest red). Single-pillar views use the pillar's color: red for exposure, orange for sensitivity, teal for adaptive capacity.

Score interpretation

Scores are *relative percentiles*, not absolute risk. A score of 75 means the tract is at higher risk than 75% of San Diego County tracts. It does not mean 75% of residents will be harmed.

Map layers

- **Composite Risk:** the 0–100 score combining all three pillars (default view).
- **Dominant Driver:** tracts colored by which pillar contributes most to the score. Distinguishes hotspots driven by exposure from those driven by sensitivity or low adaptive capacity.
- **Pillar views:** a single pillar shown alone.

Tract panel

Click a tract for its three pillar scores, its composite score, and its dominant driver.

Future scenarios

Four buttons at the top of the left panel switch the temperature variables between time periods:

- **Present Day:** observed weather-station data.
- **Moderate Emissions (SSP2-4.5):** emissions decline but not aggressively.
- **High Emissions (SSP3-7.0):** emissions remain high.
- **Very High Emissions (SSP5-8.5):** unchecked fossil-fuel use.

Future scenarios use LOCA2-downscaled CMIP6 projections for 2050 or 2080. Sensitivity and adaptive-capacity variables are held at present-day values; the tool does not project future population composition.

Examples

Identify candidate cooling-center locations

1. Start with the default selections.
2. Under **Adaptive capacity**, uncheck the AC variables so the score does not credit tracts that already have AC.
3. Click **Build Composite Risk Map**.
4. Inspect the darkest red tracts. Tracts whose dominant driver is adaptive capacity are the strongest candidates; tracts driven by exposure or sensitivity may need a different intervention.

Compare present-day and mid-century heat exposure

1. Click **High Emissions** at the top of the left panel.
2. Uncheck everything except the temperature variables.
3. Click **Build Composite Risk Map**.
4. Open a second tab on **Present Day** with the same variable selection. Differences in tract color show where projected warming is concentrated.

Limits

- **Not a real-time warning.** For active heat events, use the National Weather Service.
- **Not individual risk.** A tract score averages over residents and can hide the single block or household most exposed.
- **Quantitative only.** The score describes data patterns; it does not capture local knowledge from residents.
- **San Diego County only.** The variables come from national datasets, but the inputs must be reassembled per county. The method generalizes; the prepared data does not.
- **No save state.** The tool runs entirely in the browser with no account. Use a screenshot to capture a configuration.
- **Blank tracts indicate missing data.** A tract appears blank when one of the selected variables is missing for it. Unchecking the variable with the spottiest coverage usually fills the gaps.

To report a bug or request a feature, open an issue at github.com/connorjmack/heat-risk-viewer/issues.

Data sources

VARIABLE	SOURCE
Weather observations	MesoWest / Synoptic Data
Future temperature projections	LOCA2 downscaled CMIP6
Land surface temperature	USGS Landsat
Chronic disease prevalence	CDC PLACES
Social vulnerability	CDC SVI 2022
Income, education, healthcare, housing	Healthy Places Index (HPI)
Demographics and housing	ACS 5-year estimates
Air conditioning prevalence	Romitti et al.; Ahn et al. (modeled)

Each variable name in the tool has a small `src` link that opens the original source.